Department of Computer Science
Vrije Universiteit Amsterdam

Final Exam for the course *Logic and Sets*
March 29, 2019, 12:00–14:00

This exam has 7 pages and 7 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

Please motivate all answers!

1. **Axioms for semantic equivalence** (7 points)
Show, using the axioms for semantic equivalence, that

$$\neg\bot \;\equiv\; \top$$

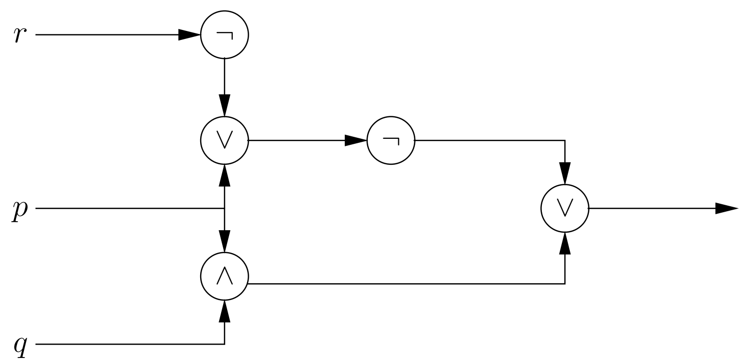
(Hint: Use the law $\phi \wedge \neg\phi \equiv \bot$*.)*

Solution: $\neg\bot \equiv \neg(\phi \wedge \neg\phi) \equiv \neg\phi \vee \neg\neg\phi \equiv \neg\phi \vee \phi \equiv \phi \vee \neg\phi \equiv \top$.

2. **Logic circuit** (7 points)
Construct a logic circuit that corresponds to the formula

$$(p \wedge q) \vee \neg(p \vee \neg r)$$

Solution:



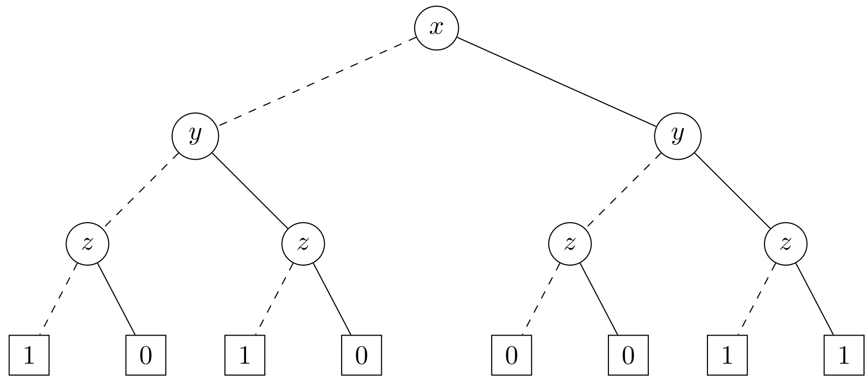
3. **OBDD** (2 + 7 + 7 points)

(a) Represent the formula

$$(\neg x \vee y) \wedge (x \vee \neg z)$$

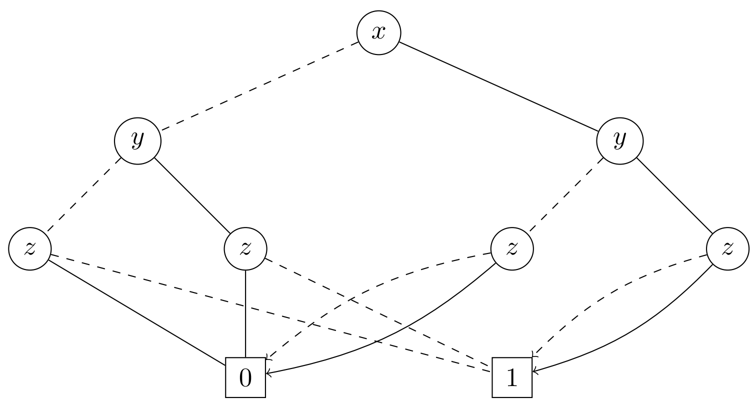
by means of a binary decision tree, with respect to the variable ordering $x,\ y,\ z$.

Solution:

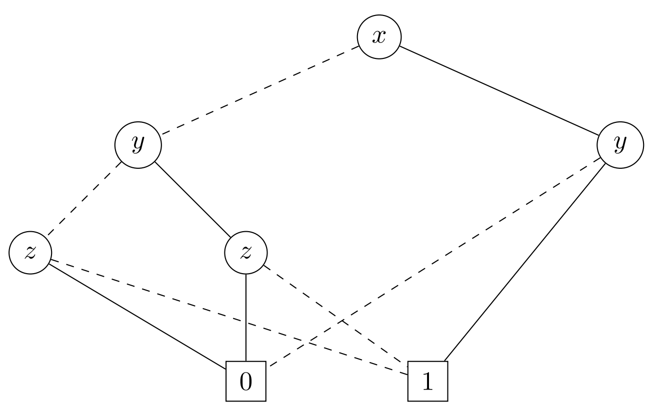


(b) Reduce the binary decision tree from (a) to a reduced OBDD, using the steps C1,2,3.

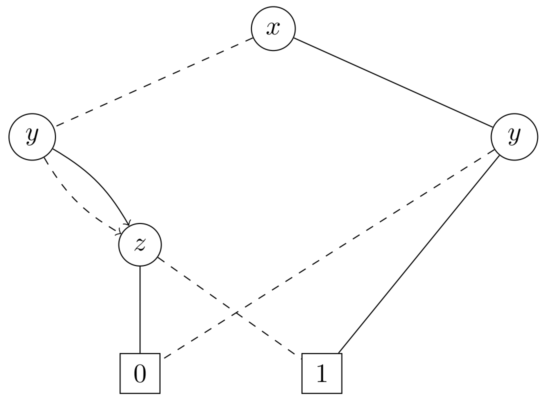
Solution: Collapse leaves (C1).



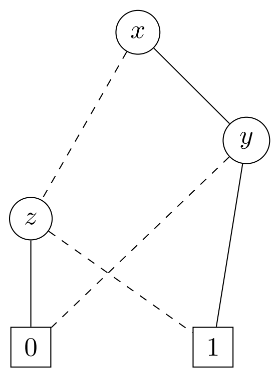
Eliminate nodes of which the 0- and 1-transition lead to the same node (C2, two times).



Collapse nodes that have identical 0-and 1-transitions.



Eliminate nodes of which the 0- and 1-transition lead to the same node (C2).

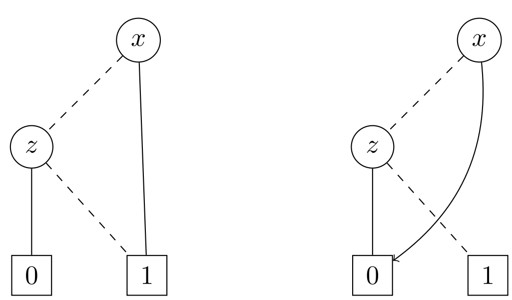


(c) Construct an OBDD for

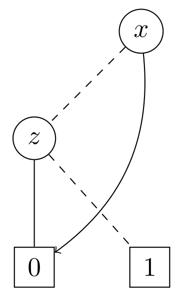
$$\forall y\,((\neg x \vee y) \wedge (x \vee \neg z))$$

from the reduced OBDD you constructed in (b).

Solution: Taking $y = 1$ and $y = 0$, we get the following two OBDDs, respectively:



To construct the OBDD for $\forall y\,((\neg x \vee y) \wedge (x \vee \neg z))$, we must compute the conjunction of these two OBDDs:



4. Predicate logic (6 + 9 points)
Consider the following two formulas:

- $\exists y \, \forall x \, R(x, y)$
- $\forall x \, \exists y \, R(x, y)$

(a) Give a model that shows these two formulas are not semantically equivalent.

Solution: One example is a model with two elements: $\{a, b\}$, with $R(a, a)$ and $R(b, b)$ as the only two relations. Then the second formula holds, because for $x = a$ we can take $y = a$ and for $x = b$ we can take $y = b$. However, since $R(a, b)$ and $R(b, a)$ do not hold, the first formula does not hold on this model. For $y = a$ the formula fails for $x = b$, while for $y = b$ the formula fails for $x = a$. (Another solution is to define $R(a, b)$ and $R(b, a)$ instead.)

(b) Does one of these formulas semantically entail the other? (Explain your answer.)

Solution: The first formula semantically entails the second one. The first formula states that there is a single element c such that for each element a we have $R(a, c)$. The second formula states that for each element a there is a (for the different a's possibly different) element c_a such that $R(a, c_a)$. If the first formula holds, then the second formula can also be fulfilled, taking $c_a = c$ for all elements a.

5. Equivalence relations (6 + 9 points)
In this problem, we consider the relation R in the set $A := \{1, 2, 3, 6, 9\} \times \{1, 2, 3, 6, 9\}$ that is defined by the description

$$\langle a, b \rangle \; R \; \langle c, d \rangle \qquad \Longleftrightarrow \qquad \frac{a}{b} = \frac{c}{d}$$

(a) Show that R is an equivalence relation.

Solution: We have to show that R is reflexive, symmetric and transitive.
Reflexivity: $a/b = a/b$ hence $\langle a,b\rangle \; R \; \langle a,b\rangle$.
Symmetry: $a/b = c/d$ implies $c/d = a/b$, hence $\langle a,b\rangle R\langle c,d\rangle$ implies $\langle c,d\rangle R\langle a,b\rangle$.
Transitivity: $a/b = c/d$ and $c/d = e/f$ together imply $a/b = e/f$, hence we also have that $\langle a,b\rangle \; R \; \langle c,d\rangle$ and $\langle c,d\rangle \; R \; \langle e,f\rangle$ together imply $\langle a,b\rangle \; R \; \langle e,f\rangle$.

(b) Explicitly write down all the different equivalence classes of the equivalence relation R in A, and give a complete system of representatives.

Solution: The equivalence classes (arranged by size) are:

$$\{\langle 1,1\rangle,\langle 2,2\rangle,\langle 3,3\rangle,\langle 6,6\rangle,\langle 9,9\rangle\}$$
$$\{\langle 1,3\rangle,\langle 2,6\rangle,\langle 3,9\rangle\} \qquad \{\langle 3,1\rangle,\langle 6,2\rangle,\langle 9,3\rangle\}$$
$$\{\langle 1,2\rangle,\langle 3,6\rangle\} \qquad \{\langle 2,1\rangle,\langle 6,3\rangle\} \qquad \{\langle 2,3\rangle,\langle 6,9\rangle\} \qquad \{\langle 3,2\rangle,\langle 9,6\rangle\}$$
$$\{\langle 1,6\rangle\} \qquad \{\langle 1,9\rangle\} \qquad \{\langle 2,9\rangle\} \qquad \{\langle 6,1\rangle\} \qquad \{\langle 9,1\rangle\} \qquad \{\langle 9,2\rangle\}$$

Hence, a complete system of representatives is for example

$$\{\langle 1,1\rangle,\langle 1,2\rangle,\langle 1,3\rangle,\langle 1,6\rangle,\langle 1,9\rangle,$$
$$\langle 2,1\rangle,\langle 2,3\rangle,\langle 2,9\rangle,\langle 3,1\rangle,\langle 3,2\rangle,\langle 6,1\rangle,\langle 9,1\rangle,\langle 9,2\rangle\}$$

NB: the equivalence classes consist of pairs $\langle a,b\rangle$ that represent the same rational number a/b.

6. Functions (5 + 5 + 5 points)
We are given the functions $sqr\colon \mathrm{R}\to\mathrm{R}$, $dec\colon \mathrm{R}\to\mathrm{R}$ and $inv\colon \mathrm{R}\to\mathrm{R}$ defined by

$$sqr(x) = x^2 \qquad dec(x) = x-1 \qquad inv(x) = \frac{1}{x}$$

(a) For each of the functions sqr, dec and inv, determine whether or not the function is injective. Please give a brief explanation of your answer in each case.

Solution: The function sqr is not injective, since e.g. $sqr(-1) = sqr(1) = 1$. The function dec is injective, because $dec(x) = x-1$ is strictly increasing in x. Finally, inv is also injective, since $1/x_1 = 1/x_2$ (with x_1 and x_2 both nonzero) implies $x_1 = x_2$.

(b) Give an explicit description (i.e. a formula for $f(x)$) of the function f defined by

$$f := inv \circ sqr^{-1} \circ dec.$$

Solution:

$$f(x) = \frac{1}{\sqrt{x-1}}$$

(c) Express the function $g\colon \mathrm{R} \to \mathrm{R}$ defined by the description

$$g(x) := \left(\frac{1+x}{x}\right)^2$$

as a composition of the functions *sqr*, *dec*, *inv* and their inverses.

Solution:

$$g = sqr \circ dec^{-1} \circ inv$$

7. Induction (5 + 10 points)

We claim that the following statement is true for all integers $n \geq 1$:

$$\sum_{k=1}^{n} 3k(k-1) = n(n^2-1)$$

(recall that $\sum_{k=1}^{n} a_k$ is shorthand notation for $a_1 + a_2 + \cdots + a_n$).

(a) Verify by explicit calculation that the statement is true for $n = 1$, $n = 2$ and $n = 3$.

Solution:

	$3n(n-1)$	$\sum_{k=1}^{n} 3k(k-1)$	$n(n^2-1)$
$n=1$	$3 \cdot 1 \cdot (1-1) = 0$	0	$1 \cdot (1^2-1) = 0$
$n=2$	$3 \cdot 2 \cdot (2-1) = 6$	$0+6=6$	$2 \cdot (2^2-1) = 6$
$n=3$	$3 \cdot 3 \cdot (3-1) = 18$	$0+6+18=24$	$3 \cdot (3^2-1) = 24$

So indeed, the statement is true for $n = 1$, $n = 2$ and $n = 3$.

(b) Using mathematical induction, prove that the statement is true for *all* integers $n \geq 1$.

Solution: We have already shown in part (a) that the statement is true for $n = 1$ (the base case), hence it only remains to do the induction step.

So let $m \geq 1$ be arbitrary, and assume as our induction hypothesis that

$$\sum_{k=1}^{m} 3k(k-1) = m(m^2-1). \qquad \text{IH}$$

It then follows that, on the one hand,

$$\begin{aligned}\sum_{k=1}^{m+1} 3k(k-1) &= \sum_{k=1}^{m} 3k(k-1) + 3(m+1)(m+1-1) \\ &\overset{\text{IH}}{=} m(m^2-1) + 3(m+1)m \\ &= m^3 - m + 3m^2 + 3m \\ &= m^3 + 3m^2 + 2m.\end{aligned}$$

On the other hand, we also have that

$$\begin{aligned}(m+1)((m+1)^2-1) &= (m+1)(m^2+2m+1-1)\\ &= m^3+m^2+2m^2+2m\\ &= m^3+3m^2+2m,\end{aligned}$$

and therefore

$$\sum_{k=1}^{m+1} 3k(k-1) = (m+1)((m+1)^2-1).$$

Hence we conclude that, for any $m \geq 1$, if the statement in our claim is true for $n = m$, then the statement is also true for $n = m+1$. Together with the base case, this proves that the statement is true for every integer $n \geq 1$.